Modelling Dynamic and Predictive Relationships Among Indian Macro-Financial Indicators: A VAR Approach

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1. INTRODUCTION

One of the critical aspects in economics and financial literature is the interdependence between a nation's macroeconomic health and the performance of its financial markets. In the global economy, this relationship is not a simple one-way path but a complex, dynamic feedback loop in which real economic shifts, price movements, and investor sentiment interact rapidly. For an emerging economy like India, understanding these linkages helps to formulate policy interventions and practice prudent risk management.

India, with its rapidly developing capital markets, has seen the BSE Sensex develop from its early, undeveloped stage into a powerful and widely recognized tool for assessing the economic health of emerging markets. The Sensex is widely considered a forward-looking proxy of corporate profitability and future economic expectations. Macroeconomic indicators like Gross Domestic Product (GDP), the aggregate measure of economic output, and the Consumer Price Index (CPI), the key metric for inflation and price stability, complement the financial dimension. The short-term interaction between these three variables, Sensex, GDP, and CPI, forms the base of monetary and fiscal policy adjustment in India.

However, traditional econometric analyses often fall short in capturing the full complexity of this trilateral relationship. Simple correlation analysis provides a static and synchronous view, ignoring the crucial effect of time and the multi-directional nature of causality. A policy shock, such as a change in the RBI's interest rate in response to rising CPI, is transmitted through various channels. To capture these intricate, lagged, and simultaneous effects, a robust multivariate time-series framework is required.
By employing advanced econometric techniques, particularly a vector Autoregression (VAR) framework, this project seeks to achieve a comprehensive understanding of the short-run dynamics and the transmission of risk. This project provides an evidence-based economic forecasting and risk assessment in India.

2. AIM OF STUDY

The primary aim of this project is to analyse and forecast the dynamic short-term interrelationships among three economic and financial indicators: BSE Sensex, the Consumer Price Index (CPI), and Gross Domestic Product (GDP). Using the standard Vector Autoregression (VAR) framework, the study seeks to map the short-run mean interactions within this tri-system.

The project has the following objectives:

- To statistically test the short-term directional influence among the three variables using the Granger Causality tests, identifying which variable's past values significantly predict another's current movement.
- To quantify the immediate and subsequent effects of unexpected shocks in one variable on the remaining two over a specified forecast horizon using Impulse Response Functions.
- To decompose the forecast error variance of each variable to determine the relative contribution of shocks originating from itself and the other two series using Forecast Error Variance Decomposition (FEVD).

3. REVIEW OF LITERATURE

To understand macro-financial dynamics, one must begin with the central puzzle — the relationship between stock returns and inflation. While the Fisher hypothesis posits that stocks should hedge against inflation, empirical studies often reveal a negative correlation. **Fama's (1981)** proxy hypothesis explains this anomaly by arguing that both inflation and stock returns respond to real economic activity (proxied by GDP). Strong growth boosts earnings and stock prices while lowering inflation, making the correlation between inflation and stocks appear negative. However, **Ram and Spencer (1983**) challenged Fama's conclusions, claiming his results relied on restrictive assumptions, keeping the stock–inflation puzzle unresolved.

Chen, Roll, and Ross (1986) expanded the debate by linking stock returns to broader macroeconomic risks such as industrial production and inflation, demonstrating that these variables represent systematic factors priced by the market. To empirically test such dynamic relationships, **Sims (1980)** introduced the **Vector Autoregression (VAR)** model, which treats all variables as endogenous and lets data reveal interdependencies without imposing strict theoretical restrictions. VAR models remain central to macroeconomic research **(Stock & Watson, 2001; 2017).**

Because macro variables like GDP and stock indices are often non-stationary, **Johansen and Juselius (1990)** developed the cointegration test to identify long-

run equilibrium relationships. If variables are cointegrated, a VECM is used; if not, a differenced VAR is appropriate. In the Indian context, **Prantik et al. (2004)** found that while neural networks outperform VARs during volatile periods, VAR models better capture the underlying economic structure, making them ideal for examining dynamic linkages.

Looking at the Empirical Evidence from the Indian Context:

Now, considering some empirical evidence specific to the Indian context. A significant body of research confirms that a relationship between macro variables and the Indian stock market exists.

For example, **Narwal and Mittal (2011)** investigated the dynamic relationship between the BSE Sensex and key macroeconomic variables, including GDP and inflation. Using the Johansen Cointegration test, they found a stable long-run equilibrium relationship between the variables. Also, their VECM analysis revealed a short-run causal relationship running from the macro variables to the stock market, giving strong evidence that macro fundamentals drive stock performance in India. Looking at a recent paper, **Chauhan et al. (2024)** corroborated these findings. Using ARDL and Granger Causality tests, they also concluded that macroeconomic variables possess predictive significance for the Indian stock market in both the short and long run. However, the empirical literature does not speak with one voice**.** In another recent study, **Rao and Jyotreddy (2022)** investigated these causal links using an ARDL approach. Their findings presented a more complex picture: while they identified a significant short-run positive relationship between inflation and stock prices, they concluded that the macroeconomic variables they studied had an insignificant impact in the long run.

The combinations of theoretical debate and, more importantly, direct empirical contradiction in the recent Indian literature, creates a clear and compelling research gap. The precise nature of the dynamic linkages, particularly the distinction between short-run dynamics and long-run equilibrium, remains an open question. This study will contribute directly to this debate. By applying a full Vector Autoregression (VAR) framework to the BSE Sensex, GDP, and inflation, this paper aims to clarify these mixed results and explicitly model the dynamic and predictive relationships between these core macro-finance indicators.

4. RESEARCH METHODOLOGY

4.1 DATA AND VARIABLES

Three variables have been used to conduct this analysis. Two of which are macroeconomic variables, and the other is a financial index.

- Macroeconomic Indicator:
  - GDP- The real GDP year-on-year growth is taken. The frequency of this data is quarterly.
  - Inflation- Consumer Price Index (CPI) is used as a metric of inflation. The year-on-year growth of CPI has been taken. The data was in monthly form, but has been averaged for each quarter.

- Financial Index:
  - BSE Sensex 30- The closing price of the index has been used. The monthly data has been averaged for each quarter.

The frequency of the data is quarterly, and the total sample period is from 2013 Q2 to 2025 Q2. Real GDP and CPI data have been sourced from CEIC Data, and BSE Sensex 30 data from Investing.com.

4.2 THEORETICAL FRAMEWORK

For the analysis, a VAR model has been chosen and estimated. The aim was to understand the interrelationships among GDP growth, Inflation growth, and the Sensex, and how they influence each other. VAR models each variable as a function of its own past and the past of all other variables in the system. This allows you to capture complex feedback loops without relying on restrictive, single-equation models.

The second objective was to model the dynamic short-term relationship of these variables. VAR is a dynamic model. It explicitly models the influence of shocks over time.

4.3 TESTS FOR STATIONARITY

The Augmented Dickey-Fuller Test has been employed to check the stationarity of the three endogenous variables. The test and the results are as follows:

H_0: Non-stationarity
H_1: Stationarity

		GDP	CPI	SENSEX
	ADF Statistic	-4.1304	-2.662	-4.09
	P-value	0.0009	0.0809	0.001
Critical Values	1%	-3.5717	-3.5848	-3.5925
	5%	-2.9226	-2.9283	-2.9315
	10%	-2.5993	-2.6023	-2.6041
	Decision	Reject Null	Fail to Reject Null	Reject Null
		Stationary	Non-stationary	Stationary

CPI is not stationary at the level. Therefore, the first difference has been taken, and stationarity has been tested on the differenced series.

		CPI First Difference
	ADF Statistic	-4.9493
	P-value	0
Critical Values	1%	-3.5848
	5%	-2.9283
	10%	-2.6023
	Decision	Reject Null
		Stationary

Thus, variables GDP and Sensex are I(0). And CPI is I(1). This justifies the use of the standard VAR model.

4.4 LAG LENGTH SELECTION

The optimal lag length is selected using Information Criteria. The following table shows the AIC, BIC, FPE, and HQIC:

VAR Lag Order Selection Results				
Lag	AIC	BIC	FPE	HQIC
0	-18.67	-18.55	7.77E-09	-18.63
1	-19.28	-18.78*	4.26E-09	-19.00*
2	-19.11	-18.23	5.07E-09	-18.79
3	-19.36	-18.11	4.01E-09	-18.91
4	-19.52	-17.89	3.580e-09*	-18.92
5	-19.35	-17.34	4.50E-09	-18.62
6	-19.39	-17.01	4.78E-09	-18.52
7	-19.5	-16.74	4.94E-09	-18.5
8	-19.95*	-16.82	3.90E-09	-18.81

The optimal lag length is chosen based on the Bayesian Information Criterion (BIC) and the Hannan-Quinn Information Criterion. Thus, the lag order is selected as 1.

5. MODEL ESTIMATION

The standard VAR has been estimated using the Ordinary Least Squares method.

Results for equation CPI First Difference				
	Coefficient	Std. Error	t-stat	P-Value
const	0.001458	0.003034	0.481	0.631
L1.CPI_diff	-0.050341	0.148526	-0.339	0.735
L1.GDP	-0.034396	0.027442	-1.253	0.21
L1.SENSEX	0.006606	0.013629	0.485	0.628

The equation being estimated is:

$$CPI_diff_t = C + \varphi_1 CPI_diff_{t-1} + \varphi_2 GDP_{t-1} + \varphi_3 SENSEX_{t-1} + \epsilon_t$$

- The average, baseline change in CPI is not statistically different from zero.
- No Persistence: The previous period's change in CPI has no significant ability to predict the current period's change. The system assumes inflation changes are independent over time in a lag 1 model.
- No Real Economy Link: Past GDP movements have no significant direct short-term impact on the change in CPI.
- No Financial Link: Past SENSEX movements have no significant direct short-term impact on the change in CPI.

Results for equation GDP				
	Coefficient	Std. Error	t-stat	P-Value
const	0.053102	0.016648	3.19	0.001
L1.CPI_diff	-0.386578	0.815018	-0.474	0.635
L1.GDP	0.344639	0.150584	2.289	0.022
L1.SENSEX	0.120526	0.074787	1.612	0.107

The equation being estimated is:

$$GDP_t = C + \varphi_1 CPI_diff_{t-1} + \varphi_2 GDP_{t-1} + \varphi_3 SENSEX_{t-1} + \epsilon_t$$

- Highly Significant Baseline: The average, baseline (autonomous) growth of GDP is strongly positive and statistically significant.

- No Monetary Link: The previous period's change in inflation CPI has no significant direct short-term impact on GDP growth.

- Significant Persistence: GDP growth is significantly and positively predicted by its own value in the last period. This shows that the momentum or inertia of GDP growth is a strong short-term factor.

- Borderline Financial Link: Past SENSEX movements have a positive influence on GDP, but this effect is not statistically significant at the 5% level (P-value is 0.107). It is significant only at the 11% level, suggesting a weak or lagged influence that VAR may not fully capture.

Results for equation SENSEX				
	Coefficient	Std. Error	t-stat	P-Value
const	0.038021	0.024814	1.532	0.125
L1.CPI_diff	-3.473150	1.214782	-2.859	0.004
L1.GDP	-0.048369	0.224444	-0.216	0.829
L1.SENSEX	0.717887	0.111470	6.440	0.000

The equation being estimated is:

$$SENSEX_t = C + \varphi_1 CPI_diff_{t-1} + \varphi_2 GDP_{t-1} + \varphi_3 SENSEX_{t-1} + \epsilon_t$$

- The baseline average return is positive but not statistically significant at the 5% or 10% level.

- Highly Significant Negative Monetary Link: The previous period's change in CPI (inflation) is a strong, statistically significant predictor of current Sensex returns. The negative coefficient of -3.47 means an unexpected increase in inflation expectations leads to a drop in the stock market.

- No Real Economy Link: Past GDP movements have no significant short-term impact on Sensex returns.

- Extremely Strong Persistence (Momentum): The most significant predictor is the Sensex's own history. The coefficient of 0.72 means about 72% of last quarter’s return momentum is carried over to the present quarter, consistent with the efficiency and liquidity of financial markets.

In the above OLS estimations, we find that GDP follows an autoregressive process where it is highly driven by its own momentum. The market index, Sensex, is strongly influenced by its own one-period lagged value and by the changes in inflation.

However, the purpose of VAR is to model the joint impact of one variable on another. Our lag length of 1 might pose a problem, as economic effects rarely manifest only in one period. Thus, we move on to check the Granger Causality test for a joint hypothesis. If this joint test is significant, the variable is a valid predictor, regardless of the individual t-tests.

6. MODEL DIAGNOSTICS

6.1 MODEL STABILITY

A stable VAR model implies that any shock to the system will decay over time, and the variables will return to their long-run average. For this, we require all the eigenvalues to be less than 1, that is, all lie inside the unit circle.

This stability check also guarantees that the impulse response functions will not show absurd, economically meaningless results. The following are the eigenvalues of our VAR(1) model:

VAR Model Stability Check
Eigenvlaues of VAR(1)
0.722696655
0.000644014
0.288843799

Here, all the eigenvalues are less than 1. Thus, the VAR(1) model is stable.

6.2 RESIDUAL SERIAL CORRELATION (PORTMANTEAU TEST)

This test checks if the error terms are white noise, that is, they have no serial correlation. If serial correlation exists, it would mean that there is still predictable information left in the error terms.

H_0: The residuals are white noise
H_1: The residuals have serial correlation

Portmanteau Test
P-Value (Portmanteau, lag 12): 0.1229

P-value is 0.1229. Since $0.1229 > 0.05$, we FAIL to Reject H_0. Thus, residuals are white noise, and the lag length is sufficient.

6.3 RESIDUAL ARCH EFFECT TEST

This test checks whether the volatility of the forecast errors is constant over time. If heteroskedasticity is present, the standard errors of the estimated coefficients would be incorrect, potentially leading to misleading t-tests and P-values.

H_0: There are NO ARCH effects. The volatility is stable and predictable (Homoskedasticity).
H_1: There ARE ARCH effects. The volatility is clustered and predictable from its own past (Heteroskedasticity).

ARCH-LM Test	
	P-value
CPI_diff	0.5651
GDP	0.2551
SENSEX	0.3066

We fail to reject H_0 in all three cases. Thus, there is no significant time-varying volatility detected in the residuals of the VAR(1) model.

7. ANALYSIS OF DYNAMIC INTERRELATIONSHIPS

7.1 GRANGER CAUSALITY TEST

Target (y)	Cause (x)	P-Value (F-Test)	Conclusion
CPI_diff	GDP	0.2123	FAIL to Reject H0 (Does NOT Granger-Cause)
CPI_diff	SENSEX	0.6287	FAIL to Reject H0 (Does NOT Granger-Cause)
GDP	CPI_diff	0.6361	FAIL to Reject H0 (Does NOT Granger-Cause)
GDP	SENSEX	0.1094	FAIL to Reject H0 (Does NOT Granger-Cause)
SENSEX	CPI_diff	0.0049	Significantly REJECT H0 (Granger-Causes)
SENSEX	GDP	0.8297	FAIL to Reject H0 (Does NOT Granger-Cause)

- We see that there is a unidirectional causal relationship running from CPI_diff to SENSEX. The past change in the Consumer Price Index contains information that can be used to statistically predict future changes in the SENSEX. This confirms that the stock market is reacting to inflation rather than leading it, which is typical for emerging markets

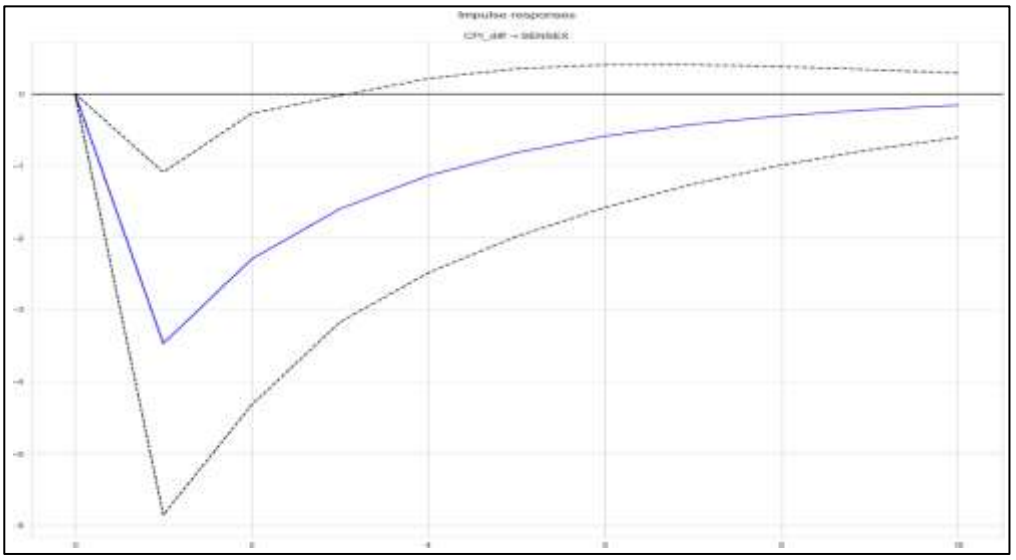
like India, where stock market movements are driven by macroeconomic fundamentals.

- SENSEX is not perfectly informationally efficient, as not all inflation data is instantly reflected in the share price. The result suggests that investors and analysts are not fully factoring in or acting on the CPI data until after it is released, giving its historical values predictive power.

7.2 IMPULSE RESPONSE FUNCTION

We will now focus on the dynamic relationship between CPI and SENSEX as these variables have significant unidirectional Granger Causation.

The Impulse Response Function (IRF) plots the effect of a shock in one variable on the current and future values of other variables in the system.



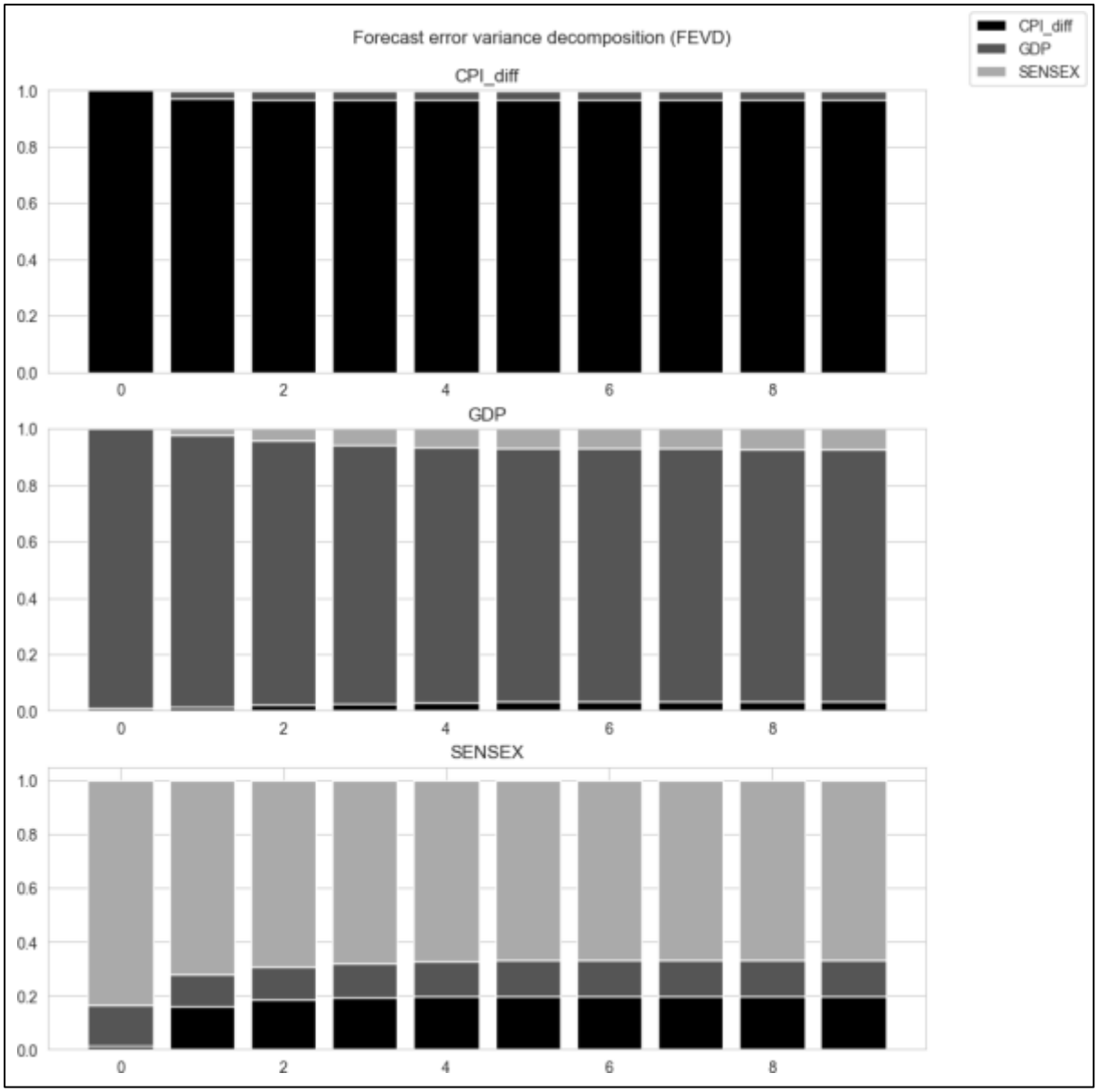
- Initial Shock and Direction (Period 0 to 1):
  The SENSEX responds immediately and negatively to the inflation shock. The blue line drops sharply from the zero line, hitting its minimum around period 1 (or slightly after).
  An unexpected increase in the rate of inflation sharply depresses the Indian stock market.

- Decay and Persistence (Period 2 onwards)
  After period 1, the blue line shows a steady, gradual recovery back toward the zero line.

The negative impact is persistent but slowly decays over the next 7-8 periods. By period 10, the effect has almost returned to zero, but still appears slightly negative. This suggests the initial shock to inflation continues to affect stock prices for a sustained period.

7.3 FORECAST ERROR VARIANCE DECOMPOSITION

The Forecast Error Variance Decomposition tells how much of the volatility in a variable is due to its own past shocks versus how much comes from other variables.



FEVD for CPI_diff			
Period	CPI_diff	GDP	SENSEX
0	1.000000	0.000000	0.000000
1	0.970771	0.026871	0.002358
2	0.967828	0.029818	0.002354
3	0.967317	0.030263	0.00242
4	0.967144	0.030361	0.002495
5	0.967063	0.030392	0.002545
6	0.967022	0.030405	0.002573
7	0.967001	0.030411	0.002589
8	0.966989	0.030414	0.002597
9	0.966984	0.030415	0.002601

- Across all periods, CPI_diff explains almost all its own forecast error variance. GDP and SENSEX explain very little (less than 1%) of CPI's variance, even after several periods.
- CPI_diff is largely driven by its own shocks, suggesting it is quite exogenous or independent in this model.
- GDP and SENSEX have minimal impact on CPI_diff. Their volatility barely explains CPI's variability even in the long run.

FEVD for GDP			
Period	CPI_diff	GDP	SENSEX
0	0.011022	0.988978	0.000000
1	0.011766	0.965719	0.022514
2	0.01982	0.936172	0.044008
3	0.026125	0.916657	0.057219
4	0.029843	0.905713	0.064444
5	0.031863	0.899877	0.06826
6	0.03293	0.896813	0.070256
7	0.03349	0.895213	0.071298
8	0.033782	0.894377	0.071841
9	0.033935	0.893941	0.072124

- At the start (period 0), GDP variance is entirely explained by its own shocks (100%). Over time, GDP continues to be dominated by its own innovations (~99%). CPI_diff and SENSEX each contribute less than 1% to GDP's forecast variance, even after several periods.
- GDP's forecast variance is almost completely explained by its own shocks, meaning GDP behaves as endogenously determined and relatively independent of CPI_diff and SENSEX.

FEVD for SENSEX			
Period	CPI_diff	GDP	SENSEX
0	0.014158	0.148758	0.837084
1	0.158308	0.118864	0.722828
2	0.183828	0.12407	0.692102
3	0.192155	0.128894	0.678952
4	0.195635	0.131697	0.672668
5	0.197271	0.133194	0.669534
6	0.198085	0.133977	0.667938
7	0.1985	0.134385	0.667115
8	0.198714	0.134598	0.666688
9	0.198826	0.134708	0.666466

- At period 0, 83.7% of SENSEX's variation is explained by its own innovations — i.e., it is mostly driven by internal or market-specific factors.
- Gradual decline in own contribution: Over time, this falls to ~66%, meaning that external macroeconomic factors start to play a slightly larger role in explaining market variance.
- Growing role of CPI_diff (inflation): The contribution of CPI_diff rises from 1.4% to nearly 20%, showing that inflation shocks

increasingly explain stock market movements over time.

- Stable but modest GDP impact: The influence of GDP remains around 13–14%, suggesting a moderate and consistent link between output and stock market fluctuations.
- Short-run: SENSEX is mainly self-driven, reflecting market-specific or internal financial dynamics.
- Medium to long-run: Inflation (CPI_diff) becomes a notable driver, implying that price level changes gradually affect investor expectations, profitability, and asset valuations.
- GDP impact: Economic growth has a steady but moderate effect, consistent with the idea that corporate earnings and output conditions influence the market but are not its sole determinant.

8. FINDINGS

1. Model Estimation Insights

- Inflation (CPI): Changes in CPI are not significantly predicted by past values of GDP or Sensex, implying that inflation dynamics are largely self-driven. CPI is an exogenous variable, reacting minimally to macro-financial shocks in the short run.
- GDP Growth: GDP growth shows significant persistence; past GDP strongly predicts future GDP. This suggests inertia in India's economic growth. There is no significant short-term effect of inflation on GDP, and only a borderline positive influence from Sensex, hinting at a weak finance-growth linkage.
- BSE Sensex: Sensex returns exhibit strong autoregressive behaviour; about 72% of momentum from the previous quarter carries over. Inflation shocks have a highly significant and negative effect on the Sensex. An unexpected rise in inflation expectations leads to a decline in stock prices, suggesting investors react negatively to inflationary pressures. GDP has no significant direct short-term impact on Sensex returns.

2. Dynamic Relationships

- Granger Causality: There exists a unidirectional causal link from inflation (CPI_diff) to Sensex, confirming that stock prices react to inflation rather than anticipate it. This implies the Indian stock market is not fully informationally efficient, as CPI data releases affect prices with a lag.

- Impulse Response Function (IRF): A positive shock in inflation causes an immediate and sharp fall in the Sensex, reaching its trough around one quarter after the shock. The negative impact persists for 8–10 periods before gradually decaying — evidence of prolonged market sensitivity to inflationary pressures.
- Forecast Error Variance Decomposition (FEVD):
    - CPI_diff: Almost entirely explained by its own shocks, confirming its exogeneity.
    - GDP: Nearly 99% of its forecast variance comes from its own shocks — it remains internally driven.
    - Sensex: Initially 83.7% self-driven; over time, the influence of inflation rises to about 20%, while GDP explains around 13–14%. This indicates that while market factors dominate in the short run, inflation becomes a significant driver in the medium to long run.

3. Overall Interpretation

- Inflation shocks are **the key external macroeconomic driver** affecting India's stock market.
- GDP growth, while persistent, has a **limited immediate impact** on financial market movements.
- The Indian market shows **partial inefficiency**, adjusting gradually to macroeconomic news, particularly inflation.

9. CONCLUSION

This study set out to explore how India's key economic indicators — GDP growth, inflation, and the BSE Sensex — interact and influence one another over time. Using a VAR model allowed us to see these relationships without assuming any one variable drives the others. The findings show that both GDP and inflation tend to move mainly on their own, while the stock market is strongly shaped by its own momentum and by changes in inflation. A sudden rise in inflation leads to a noticeable and lasting fall in the Sensex, suggesting that higher prices dampen investor confidence and reduce market returns.

The analysis also showed that inflation tends to drive stock market movements, not the other way around. This means the Indian market reacts to inflation data rather than anticipating it, implying that information takes time to be fully

reflected in prices. GDP growth, on the other hand, displays strong persistence but has little short-term impact on market performance.

Overall, the study highlights that inflation plays a key role in shaping stock market behaviour in India. When inflation rises unexpectedly, it affects investor sentiment and market valuations for several quarters. Maintaining stable prices, therefore, becomes essential not only for economic growth but also for market stability. For investors and policymakers alike, these results emphasize the importance of tracking inflation trends closely, as they remain one of the strongest signals for how the market is likely to move in the near future.

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